Set 26: Student Performance Analysis

AIM

To analyze student performance using R programming by creating a histogram of Mathematics scores, a boxplot of Science scores by gender, a scatter plot of study hours versus Mathematics scores, and a time series chart of monthly average Mathematics scores.

ALGORITHM

1. Create a dataset containing Student ID, Gender, Study Hours, Attendance, Mathematics Score, Science Score, and Exam Date.
2. Load the ggplot2 and dplyr libraries.
3. Create a histogram to display the distribution of Mathematics scores.
4. Create a boxplot to compare Science scores by gender.
5. Create a scatter plot showing the relationship between study hours and Mathematics scores, using different colors for gender and adding a regression line.
6. Convert Exam Date into date format and calculate monthly average Mathematics scores.
7. Plot the monthly averages with a moving-average smoothing line.
8. Interpret the results and identify the trends in student performance.

PROGRAM

```
library(ggplot2)

library(dplyr)


student <- data.frame(

  Student_ID = c("S01", "S02", "S03", "S04", "S05", "S06"),

  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),

  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),

  Attendance = c(78, 90, 70, 95, 85, 92),

  Math_Score = c(62, 80, 55, 90, 72, 82),

  Science_Score = c(65, 85, 58, 92, 74, 86),

  Exam_Date = as.Date(c("2025-01-10", "2025-01-10",

                        "2025-02-12", "2025-02-12",

                        "2025-03-15", "2025-03-15"))

)
```

```
ggplot(student, aes(Math_Score)) +
  geom_histogram(bins = 5, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Mathematics Scores",
       x = "Math Score", y = "Frequency")

ggplot(student, aes(Gender, Science_Score, fill = Gender)) +
  geom_boxplot() +
  labs(title = "Science Scores by Gender",
       x = "Gender", y = "Science Score")

ggplot(student, aes(Study_Hours, Math_Score, color = Gender)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Study Hours vs Mathematics Scores",
       x = "Study Hours", y = "Math Score")

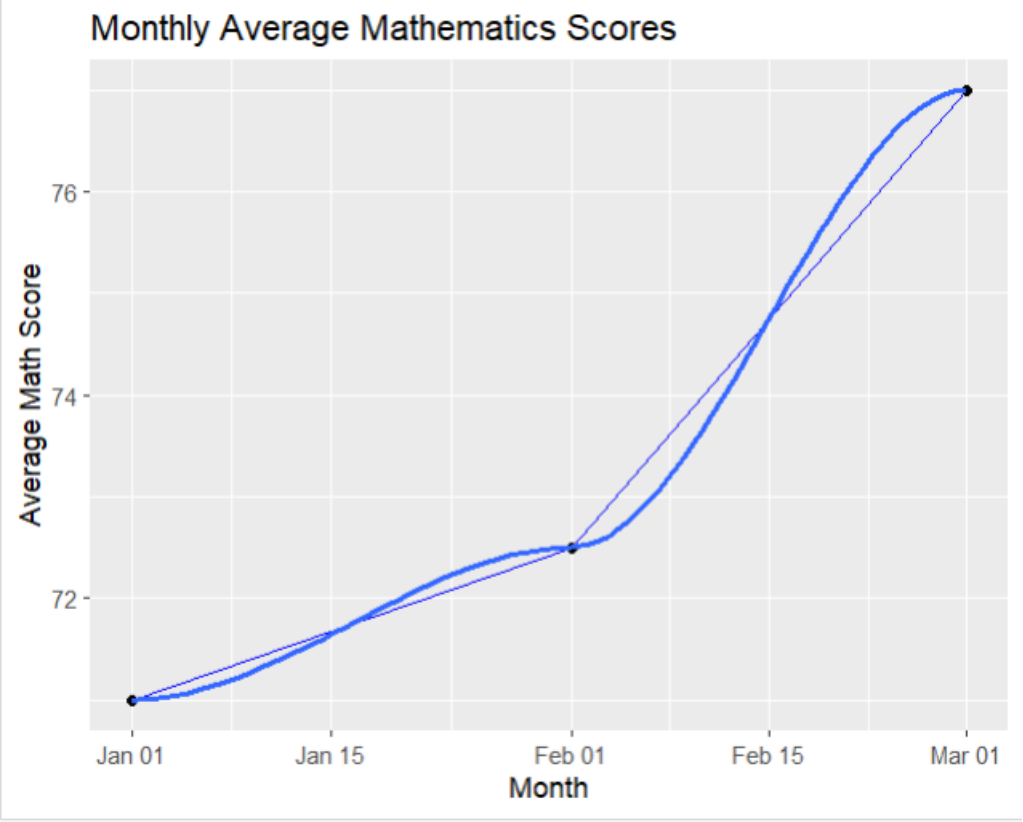
monthly <- student %>%
  mutate(Month = as.Date(format(Exam_Date, "%Y-%m-01"))) %>%
  group_by(Month) %>%
  summarise(Avg_Math = mean(Math_Score), .groups = "drop") %>%
  arrange(Month)

ggplot(monthly, aes(Month, Avg_Math)) +
  geom_line(color = "blue") +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(title = "Monthly Average Mathematics Scores",
       x = "Month", y = "Average Math Score")
```

print(student)

print(monthly)

OUTPUT:



RESULT

The histogram displays the distribution of Mathematics scores. The boxplot compares Science scores between male and female students. The scatter plot helps examine the relationship between study hours and Mathematics performance, while the monthly trend chart displays changes in average Mathematics scores over time.

The average Mathematics scores are:

Month	Average score
January 2025	71
February 2025	72.5
March 2025	77

The average Mathematics score increases from January to March, indicating an improving trend in this dataset.